

TANKER FLEET MANAGEMENT | DAHEJ LNG TERMINAL

From Manual Queues to Manless Movement

A self-service, FIFO-driven tanker ecosystem connecting requisition, approval, identity verification, token allocation, bay direction and exit closure.

DESIGN CAPACITY**100 Movements / Day****LOADING EXPANSION****4 to 8 Bays****CORE PRINCIPLE****Transparent FIFO**

Fair allocation. Faster movement. Complete digital traceability.

Installed self-service kiosk, face verification, QR/code entry, token printing and live bay display.

Project reference: Petronet LNG Composite Order 4400000672, dated 10 September 2024.

The Operating Problem to Solve

CLIENT SITUATION

Growing tanker demand was outgrowing email-led coordination.

The terminal was filling about 45 LNG trucks per day. Demand was expected to approximately double while loading capacity expanded from four bays to eight.

EXISTING THROUGHPUT

Approx. 45 / Day

EXPECTED GROWTH

Approx. 2X

PLANNED BAYS

8 Loading Bays

The manual pain points

Queue fairness

Manual coordination can create disputes about tanker sequence and priority.

Identity assurance

The vehicle, approved driver and loading request needed to match at arrival.

Email-dependent documents

Registration, passes, inspections and changes moved across departments by email.

Limited live visibility

Customers, security and operations needed a common view of queue position and movement.

What the Automated TFM Had to Deliver

A digital process from customer request to tanker exit.

The requirement combined workflow software, controlled access, self-service hardware, document storage, field infrastructure and operational support.

01

DIGITAL REQUISITION

Customer requests for loading, new tanker, new driver, gate pass, haulier change and quantity.

02

MULTI-LEVEL APPROVAL

Master, Safety and Security stages with approval, rejection and additional-document handling.

03

UNIQUE ACCESS CODE

Individual QR code and four-digit code for every approved tanker request.

04

SELF-SERVICE ARRIVAL

QR/code entry, driver face verification and automatic thermal token printing.

05

TRANSPARENT FIFO

First-in-first-out sequence, live position display and allocation across loading bays.

06

ENTRY / EXIT CLOSURE

Scan-based in/out punching; exiting tanker closes its token and advances the queue.

07

WEB + CLOUD PLATFORM

Role-based web access, five-year licence/storage and controlled data retention.

08

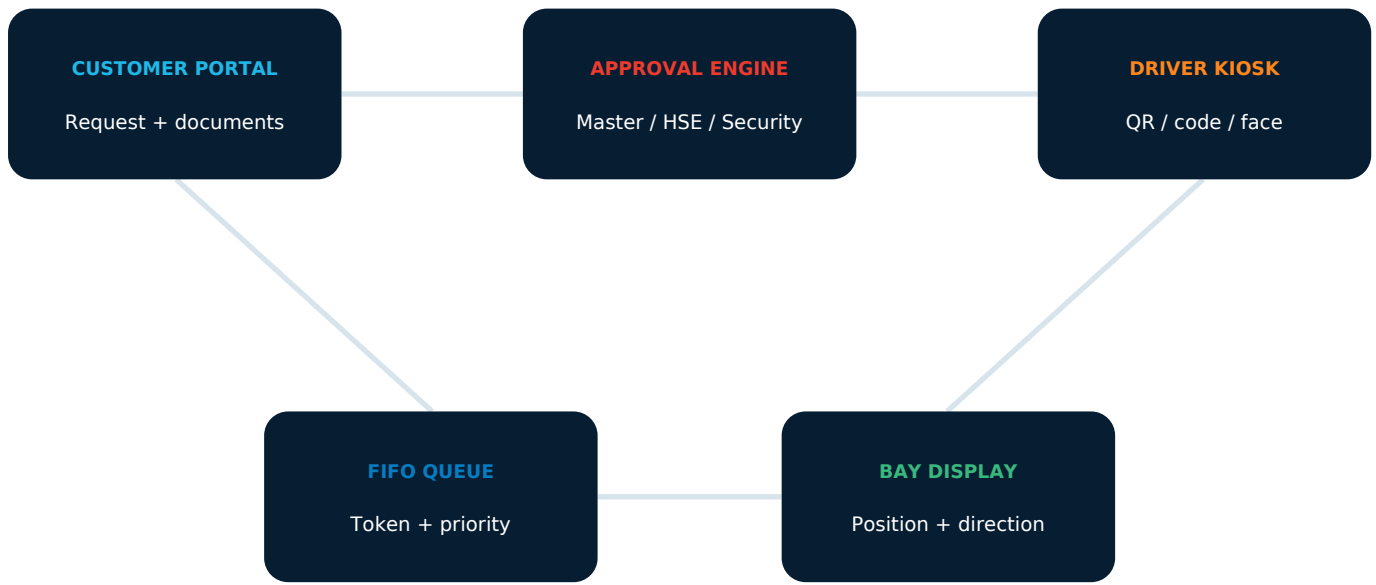
REPORTING & AUDIT

Real-time movement view and daily reports for operational traceability.

Manless Front-End, Governed Back-End

Automation at the gate. Control inside the workflow.

Manless does not mean uncontrolled. Drivers use the self-service touchpoint while PLL teams retain role-based approval, safety, security and master control.



CORE AUTOMATION LAYERS

● Identity layer

Customer, tanker, haulier and driver records with supporting documents.

● Decision layer

Configured roles, approval paths, rejection handling and access rights.

● Transaction layer

Requisition, token, arrival, queue position, bay call and exit status.

● Evidence layer

Time stamps, scanned documents, movement history, alerts and reports.

CONTROLLED AUTOMATION

The system automates repeatable steps while preserving human approval for safety and governance.

The Automated Tanker Journey

One approved request becomes one traceable movement.

The journey is designed to remove counter dependency, prevent queue ambiguity and keep every stage visible to authorized stakeholders.

01**REQUEST**

Customer submits next-day loading demand and required tanker/driver documents.

02**APPROVE**

PLL roles review the request, safety status, gate pass and loading slot.

03**AUTHORIZE**

Approved tanker receives a unique QR code and four-digit code.

04**VERIFY**

At arrival, the driver scans/enters the code and completes face verification.

05**PRINT TOKEN**

The kiosk automatically prints the tanker token and FIFO position.

06**CALL TO BAY**

Live display directs the correct tanker toward the available loading bay.

07**LOAD & MOVE**

Operations process the tanker while its status remains visible in the portal.

08**EXIT & CLOSE**

Exit scan records outward movement, closes the token and advances the next tanker.

Role-Based Portals and Data Control

The right action, by the right role, with a digital record.

The specification defines five user experiences so customers can request, while PLL teams review, approve, monitor and govern the process.

MASTER / SIC	Approvals, rejection, slots, tanker data, driver training flow, gate pass, haulier change.
CUSTOMER	Tanker linking, documents, driver requests, daily quantity, token status and change requests.
SAFETY / HSE	Driver safety-training approval and new-tanker inspection approval.
SECURITY	Tanker database, gate-pass information, approved requisitions and in/out visibility.
GUEST	Daily requisition and permitted token/movement visibility.

CONTROLLED USERS 20 Logins	MASTER DATA 500 Tankers	DRIVER PROFILES 500 Drivers
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Document capacity includes supporting scans/photos; daily movement design basis is 100 tankers.

The Physical Self-Service Solution

Digital workflow made visible at the tanker touchpoint.

The supplied project photograph documents the external TFM facility, eight-position loading display and self-service interaction equipment.



SPECIFIED FIELD COMPONENTS

Identity and access

IP facial-recognition camera, QR scanner and keypad, including specified spares.

Live operations display

Customizable 50-inch token display for queue and bay information.

Outdoor integration

Weatherproof enclosure, frame/canopy and equipment mounting.

Self-service output

Automatic network thermal token printer, with spare.

Network and power

6U rack, switch, patching/LIU, UPS with minimum two-hour backup.

Site connectivity

Armoured power and fibre cable, trenching, GI pipe and UPVC routing.

Actual installed-site photograph supplied for this success story.

Key Benefits and Success Value



PETRONET LNG | DAHEJ LNG TERMINAL

A fair, scalable and auditable tanker movement ecosystem.

The result converts disconnected emails and manual queue coordination into a governed digital process with self-service execution at the driver touchpoint.

QUEUE LOGIC

FIFO Transparency

TOUCHPOINT

Self-Service

DATA RETENTION

5 Years

Key benefits

● Fair and transparent allocation

FIFO removes ambiguity and gives customers a visible basis for tanker sequence.

● Reduced manual dependency

Drivers can verify, generate and print tokens without routine counter handling.

● Faster tanker turnaround

Digital approval and live bay direction reduce avoidable waiting and coordination delays.

● Stronger identity control

Approved tanker, driver face, token and movement record are linked in one process.

● Cross-department workflow

Customer, Operations, HSE and Security work through controlled roles and common data.

● Audit-ready evidence

Documents, time stamps, in/out events, status history and reports improve traceability.

EVIDENCE BASIS

Petronet LNG Composite Order 4400000672, Annexure-B scope, technical requirements and installed-site photograph.

D-TECH | AUTOMATION THAT MOVES OPERATIONS FORWARD